

Biometric Event Server -- Setup Guide

Target hardware: Orange Pi (4-core 32-bit ARM, Armbian, 1GB RAM, 16GB SD)

1. Prerequisites

SSH into your Orange Pi, then run:

```
sudo apt update && sudo apt upgrade -y  
sudo apt install python3 python3-venv python3-pip sqlite3 -y
```

Verify Python version (3.7+ required):

```
python3 --version
```

2. Create Project User and Directory

Running as a dedicated user limits blast radius if something goes wrong.

```
sudo useradd -r -m -s /bin/false biometric  
sudo mkdir -p /opt/biometric/data /opt/biometric/logs
```

3. Copy Project Files

From your development machine, copy the project files to the Orange Pi:

```
scp -r /path/to/biometric/* user@<ORANGE_PI_IP>:/tmp/biometric/
```

Then on the Orange Pi, move them into place:

```
sudo cp -r /tmp/biometric/* /opt/biometric/  
sudo chown -R biometric:biometric /opt/biometric
```

Or if you have git set up:

```
sudo -u biometric git clone <your-repo-url> /opt/biometric
```

4. Configure Device IPs

Edit `config.py` to match your actual ZKTeco devices:

```
sudo nano /opt/biometric/config.py
```

Update the DEVICES list:

```
DEVICES = [  
    {"id": "device_1", "ip": "192.168.1.201", "port": 4370, "password": 0, "timeout": 5},  
    {"id": "device_2", "ip": "192.168.1.202", "port": 4370, "password": 0, "timeout": 5},  
]
```

Note: `password` is `0` for most ZKTeco devices unless you explicitly set one. `port` is almost always `4370`.

5. Create Python Virtual Environment and Install Dependencies

```
sudo -u biometric python3 -m venv /opt/biometric/venv  
sudo -u biometric /opt/biometric/venv/bin/pip install --upgrade pip  
sudo -u biometric /opt/biometric/venv/bin/pip install pyzk
```

Verify pyzk installed correctly:

```
sudo -u biometric /opt/biometric/venv/bin/python -c "from zk import ZK; print('pyzk OK')"
```

6. Test Device Connectivity

Before starting the server, confirm you can reach each device:

```
sudo -u biometric /opt/biometric/venv/bin/python -c "  
from zk import ZK  
zk = ZK('192.168.1.201', port=4370, timeout=5)  
conn = zk.connect()  
print('Device name:', conn.get_device_name())  
conn.disconnect()  
"
```

Repeat for `192.168.1.202`. If this fails, check:

- Device is powered on and connected to the same LAN
- IP address is correct (check device menu: Menu ? Comm ? Ethernet)

- No firewall blocking port 4370 between the Pi and the device

7. Run Manually to Verify

Before installing as a service, do a quick sanity check:

```
cd /opt/biometric
sudo -u biometric /opt/biometric/venv/bin/python main.py
```

You should see log output like:

```
2026-04-10 12:00:00 INFO      main: Biometric Event Server starting
2026-04-10 12:00:00 INFO      device: Connected to device_1 (192.168.1.201:4370)
2026-04-10 12:00:00 INFO      device: Connected to device_2 (192.168.1.202:4370)
2026-04-10 12:00:00 INFO      api: API server listening on port 8000
```

Punch a finger on a device, then in another terminal:

```
sqlite3 /opt/biometric/data/events.db "SELECT * FROM events ORDER BY id DESC LIMIT 5;"
```

Press Ctrl+C to stop. Confirm the log shows "Shutdown complete" and no events were lost.

8. Install as systemd Service

```
sudo cp /opt/biometric/biometric.service /etc/systemd/system/
sudo systemctl daemon-reload
sudo systemctl enable biometric
sudo systemctl start biometric
```

Check it started correctly:

```
sudo systemctl status biometric
```

View live logs:

```
journalctl -u biometric -f
```

9. Install Cloudflare Tunnel

9.1 Install cloudflared (32-bit ARM build)

```
wget https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-linux-arm
chmod +x cloudflared-linux-arm
sudo mv cloudflared-linux-arm /usr/local/bin/cloudflared
cloudflared --version
```

9.2 Authenticate and Create Tunnel

```
cloudflared tunnel login
```

A browser link will print -- open it on another machine and authorize your domain.

```
cloudflared tunnel create biometric
```

Note the **Tunnel ID** printed (a UUID like xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx).

9.3 Create Tunnel Config

```
sudo mkdir -p /etc/cloudflared
```

```
sudo tee /etc/cloudflared/config.yml <<EOF
tunnel: <YOUR_TUNNEL_ID>
credentials-file: /root/.cloudflared/<YOUR_TUNNEL_ID>.json

ingress:
  - hostname: biometric.yourdomain.com
    service: http://127.0.0.1:8000
  - service: http_status:404
EOF
```

Replace <YOUR_TUNNEL_ID> and biometric.yourdomain.com with your actual values.

9.4 Add DNS Record

```
cloudflared tunnel route dns biometric biometric.yourdomain.com
```

9.5 Install and Start cloudflared Service

```
sudo cloudflared service install
sudo systemctl enable cloudflared
sudo systemctl start cloudflared
sudo systemctl status cloudflared
```

9.6 Set Access Policy (Cloudflare Dashboard)

- Go to **Zero Trust ? Access ? Applications ? Add an application**
- Choose **Self-hosted**
- Set the domain to `biometric.yourdomain.com`
- Add an **Allow** policy -- e.g., email OTP for your email address
- Save

After this, visiting `https://biometric.yourdomain.com` will require authentication before reaching the API.

10. Verify Everything End-to-End

```
# Service health
sudo systemctl status biometric
sudo systemctl status cloudflared

# Local API check
curl http://127.0.0.1:8000/status

# Remote API check (after Cloudflare auth)
curl https://biometric.yourdomain.com/status

# Latest events
curl "https://biometric.yourdomain.com/events?limit=5"

# Events by user
curl "https://biometric.yourdomain.com/events?user_id=123"

# Events from a date
curl "https://biometric.yourdomain.com/events?from=2026-04-10"
```

11. Ongoing Maintenance

View logs

```
# Live service log
journalctl -u biometric -f

# Rotating file log
tail -f /opt/biometric/logs/server.log
```

Check database size

```
du -sh /opt/biometric/data/events.db
sqlite3 /opt/biometric/data/events.db "SELECT COUNT(*) FROM events;"
```

Restart after config change

```
sudo systemctl restart biometric
```

Update code

```
sudo systemctl stop biometric
sudo cp new_files /opt/biometric/
sudo chown -R biometric:biometric /opt/biometric
sudo systemctl start biometric
```

Monitor memory usage

```
ps aux | grep python
# Expect ~30-50MB RSS total
```

Troubleshooting

Symptom	Likely cause	Fix
`Connected` then immediately `disconnected`	Wrong IP or device off	Check device IP in menu: Comm ? Ethernet
`ZKNetworkError` in logs	Firewall or VLAN blocking port 4370	Verify network path, check `ping <device_ip>` from PI
No events after punching	Device in wrong mode	Ensure device is not in "enrollment" mode
API returns 502 via Cloudflare	`biometric` service not running	`sudo systemctl start biometric`
`cloudflared` fails to start	Wrong tunnel ID or missing credentials file	Re-run `cloudflared tunnel login` and recreate config
SD card filling up	Logs or DB growing	Check `du -sh /opt/biometric/data /opt/biometric/logs`